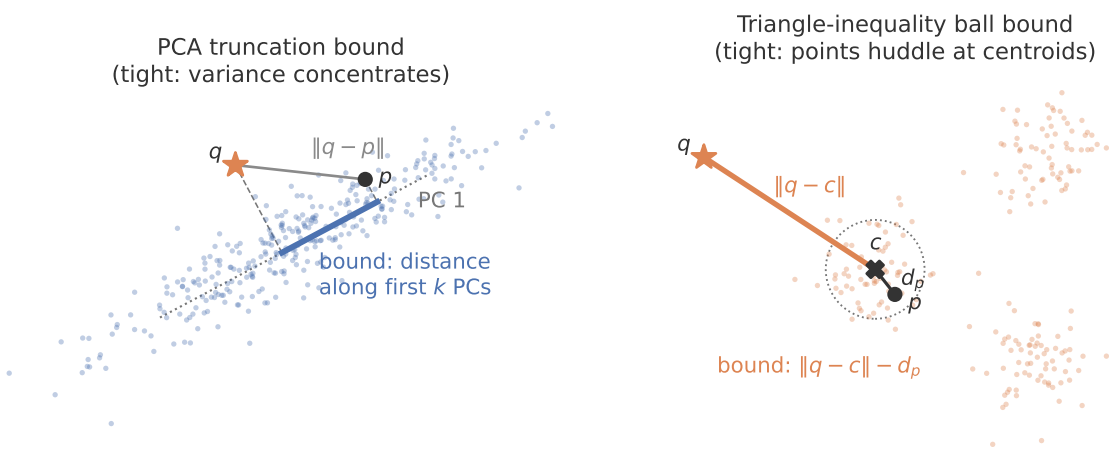
PCA truncation bound
(tight: variance concentrates)
q
‖q − p‖
p
PC 1
bound: distance
along first k PCs
Triangle-inequality ball bound
(tight: points huddle at centroids)
q
‖q − c‖
c
d_p
p
bound: ‖q − c‖ − d_p